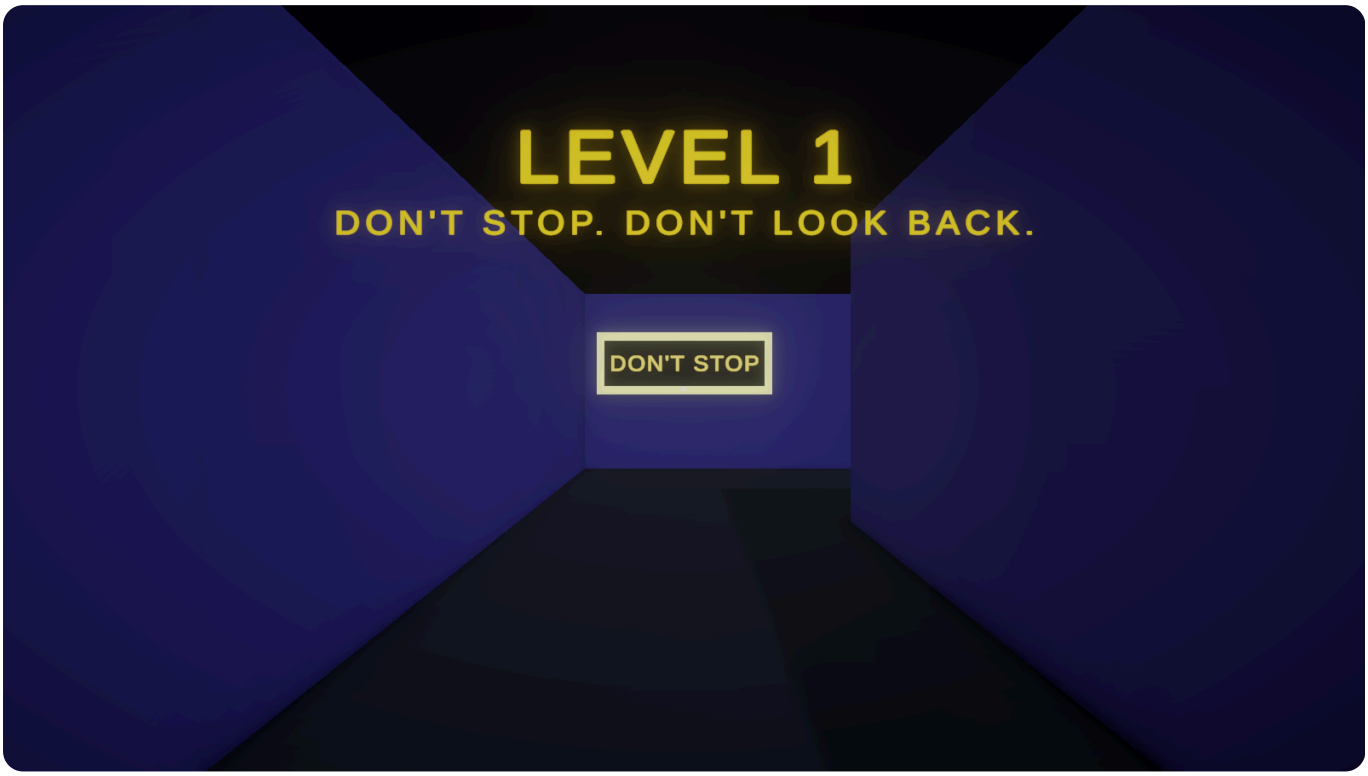
Game Design Document

Sprint or Shift

A first-person maze runner where you cannot stop, and the maze changes behind your back.



Genre: first-person maze / action-puzzle | Engine: Unity 6 (URP) | Platform: PC, keyboard + mouse (macOS build) | Session: 3–8 minutes | Audience: 12+, casual-to-core players

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Course: Game Design
Instructor: Dr. Arafa Taher
Date: October 2026

Repository: github.com/adel-nabih/SprintOrShift

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1. Overview

Elevator pitch. You are in a dark neon maze. If you stand still for two seconds, you are dragged back to the start. Every time you turn your back on a purple wall, the maze rearranges itself. You can't stop and you can't go back. You can only keep moving until you reach the glowing portal.

Design question the prototype tests

> Can two simple restrictions, **"don't stop"** and **"don't look back"**, turn an ordinary maze into a tense experience where players must commit to their choices?

Design pillars

Pillar	What it means for design decisions
Momentum over caution	Every rule rewards moving and punishes hesitating. There is no pause, no map and no slow "careful" option.
The maze is watching	The world reacts to where you look. Walls change only behind your back, so looking (or not looking) is part of how you play.
Readable rules, chaotic results	Each rule is simple and stated on a wall sign. The chaos comes from how the rules combine, not from randomness.
Small and focused	Two levels that test the idea well, instead of a large game that isn't finished.

Player experience goals

- **Urgency:** the player's heart rate goes up the moment they stop.
- **Paranoia:** players think twice before turning around.
- **Mastery:** players discover that standing still is also a deliberate way out of a dead end.

2. Core rules & controls

1. **Hesitation penalty: "DON'T STOP".** If your horizontal speed stays below 0.1 m/s for **2.0 s**, you are teleported to the START point. A red overlay with a countdown fades in from 0.3 s to warn you. The timer only starts after your first step, so waiting at START is safe.
2. **Wall shift: "DON'T LOOK BACK".** Purple walls change only after you have *seen* them (on screen with a clear line of sight) *and* they are now *behind* you (more than 110° from where you're looking). One-way **gates** close after you pass through them. **Toggle walls** flip open or closed each time.
3. **Dead ends & reset.** Dead ends are deliberate traps. If you're boxed in, standing still for 2 s sends you back to START, and **every purple wall resets** to its starting state, so the level is always solvable.
4. ★ **Win condition.** Touch the glowing cyan Finish Portal. Level 1 leads to Level 2, which leads to the Victory screen (press **R** to play again).

Input	Action	Input	Action
W A S D / arrows	Move (smooth acceleration, 6.5 m/s)	Mouse	Look (pitch clamped to ±85°)
Esc	Release the mouse cursor	Left click	Capture the mouse again
R	Restart (Victory screen only)		

3. Core loop

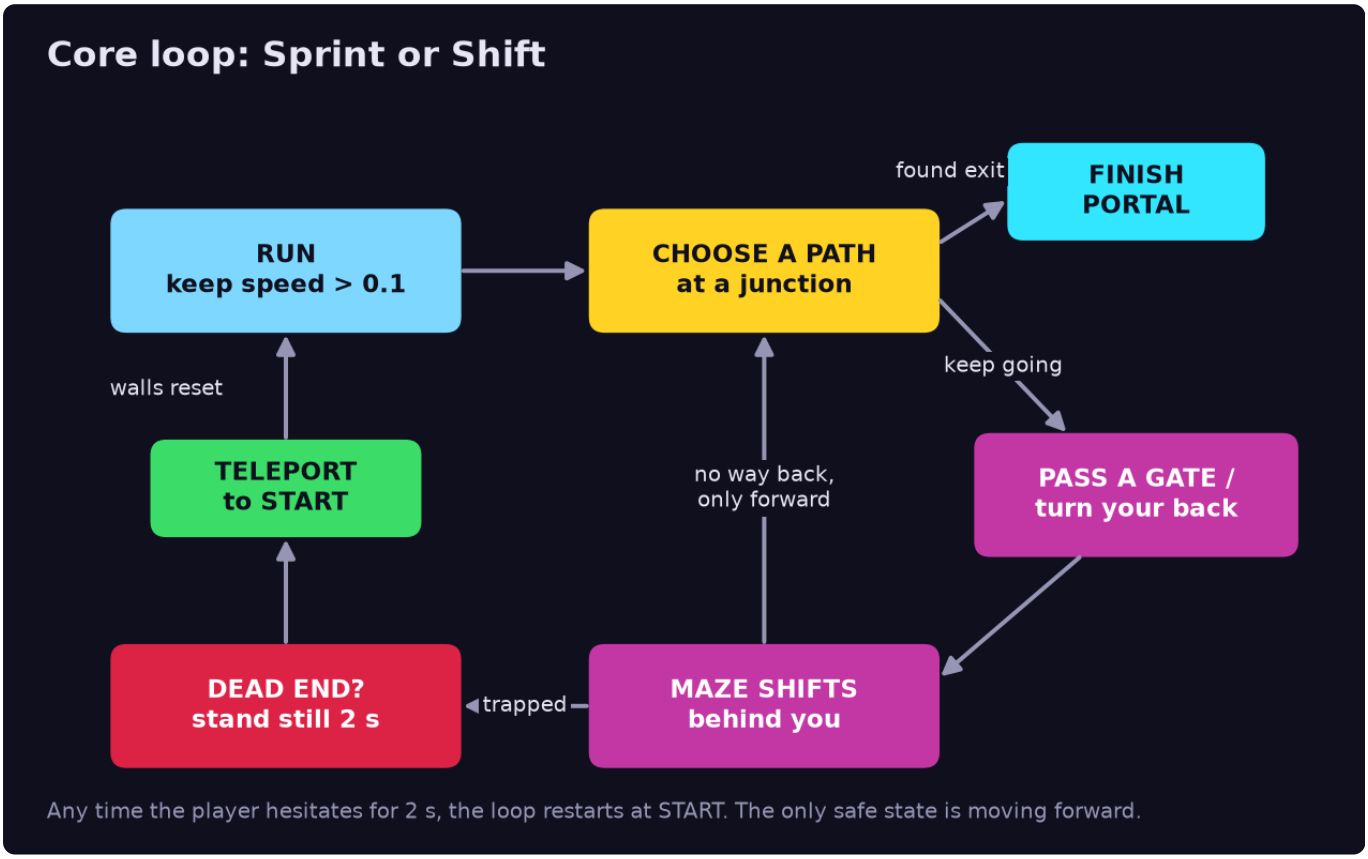


Figure 1. The core loop. Every path either moves forward (towards the portal) or ends with a reset at START. There is no safe resting state.

Second to second, the player **runs** and **chooses at junctions** under the 2-second pressure. Minute to minute, they **learn the maze** across resets: each teleport costs time but teaches them the layout and where the traps are. Across the game, Level 1 introduces each rule on its own, and Level 2 combines them all with faster walls.

4. Mechanics in detail

4.1 Wall-shift state machine

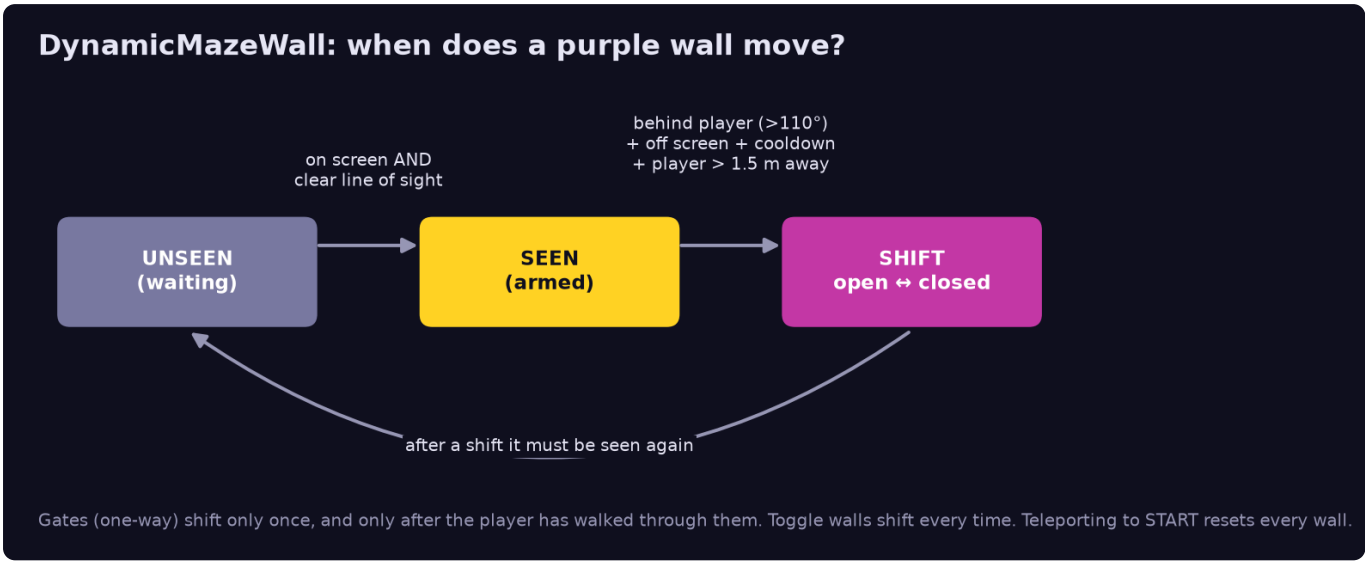


Figure 2. When a purple wall moves. The "seen" check uses the camera's view frustum *and* a line-of-sight test, so walls hidden behind other walls never count as seen.

Wall type	Starts	Shifts when…	Purpose
One-way gate (cyan arrow on the maps)	Open	Seen, the player has walked ≥ 3 m past it in the arrow direction, and it is now behind them. Shifts once only.	Stops the player retracing their steps and turns wrong choices into dead ends.
Toggle wall (⇄ on the maps)	Open	Seen, then behind the player. Flips every time. Cooldown is 2.0 s in L1 and **0.6 s** in L2.	Changes the paths: creates shortcuts or seals off a region.

Safety rules: a wall never shifts while the player is within 1.5 m of it, so it can't trap them inside the wall. Every wall resets when the player is teleported.

4.2 Stillness rule tuning

Parameter	Value	Why
Stillness limit	2.0 s	Long enough to turn a corner and read a sign. Too short to stop and study the maze.
Speed threshold	0.1 m/s	Uses the speed the player *actually* moves at, so pushing into a wall counts as standing still. This is what makes Rule 3 work.
Warning overlay	Fades in from 15%–60% of the limit	Ignores tiny pauses, then makes the danger obvious.

5. Level design

Both levels use 4 m grid cells. They are written as text layouts in `SprintOrShiftLevelBuilder.cs`, so designers can edit a maze by typing characters. Both layouts were checked with a breadth-first-search solver: the exit can be reached even when every toggle wall is closed.

Level 1: Tutorial ("DON'T STOP. DON'T LOOK BACK.")

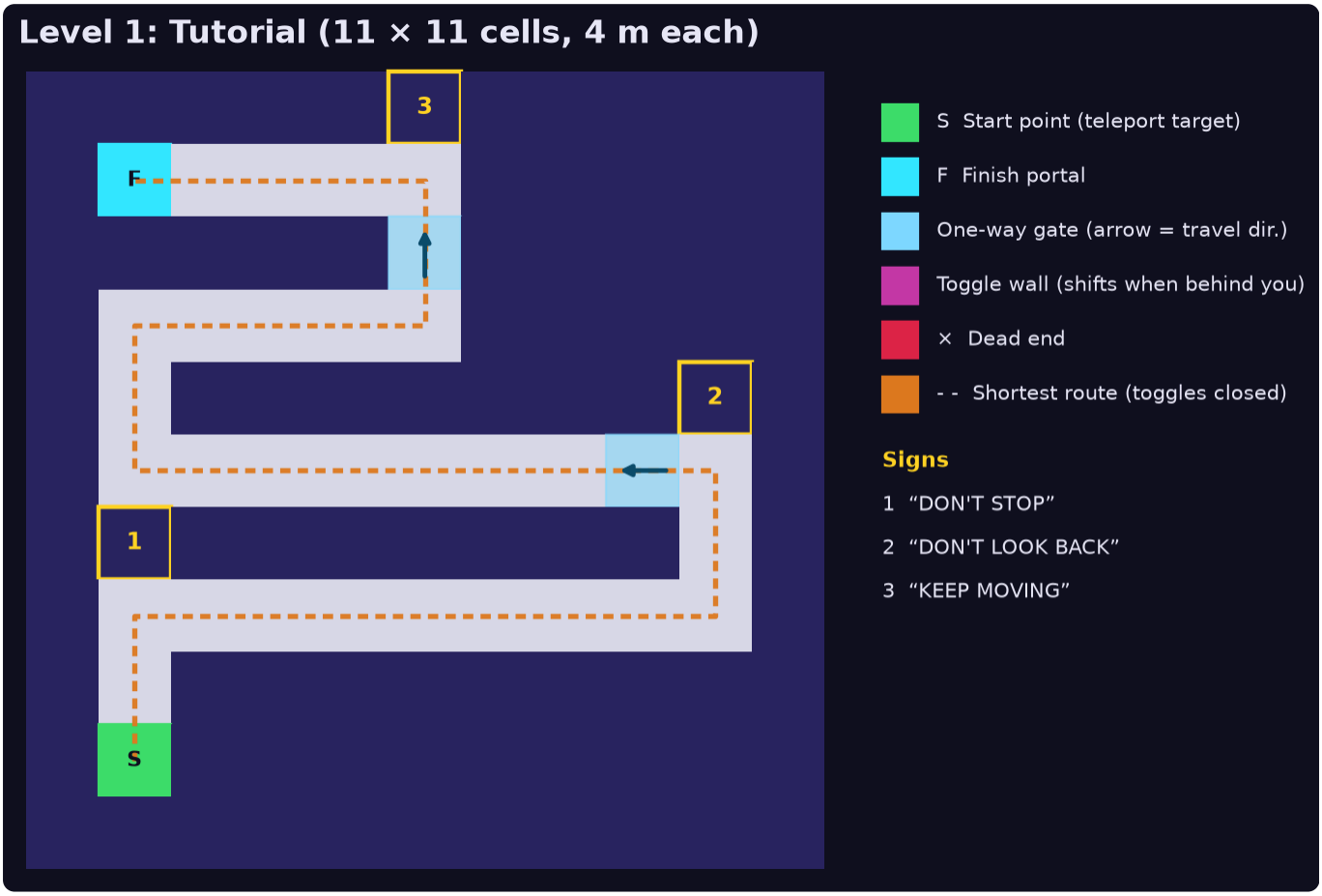


Figure 3. Level 1: one serpentine corridor (33-cell route) with 2 one-way gates and 3 signs. No branches, so the player learns each rule without getting lost.

Teaching beats: (1) the first thing the player sees is "DON'T STOP". (2) "DON'T LOOK BACK" is placed just before the first gate, which closes behind them a few seconds later. (3) "KEEP MOVING" is placed before the second gate and the portal.

Level 2: High Chaos

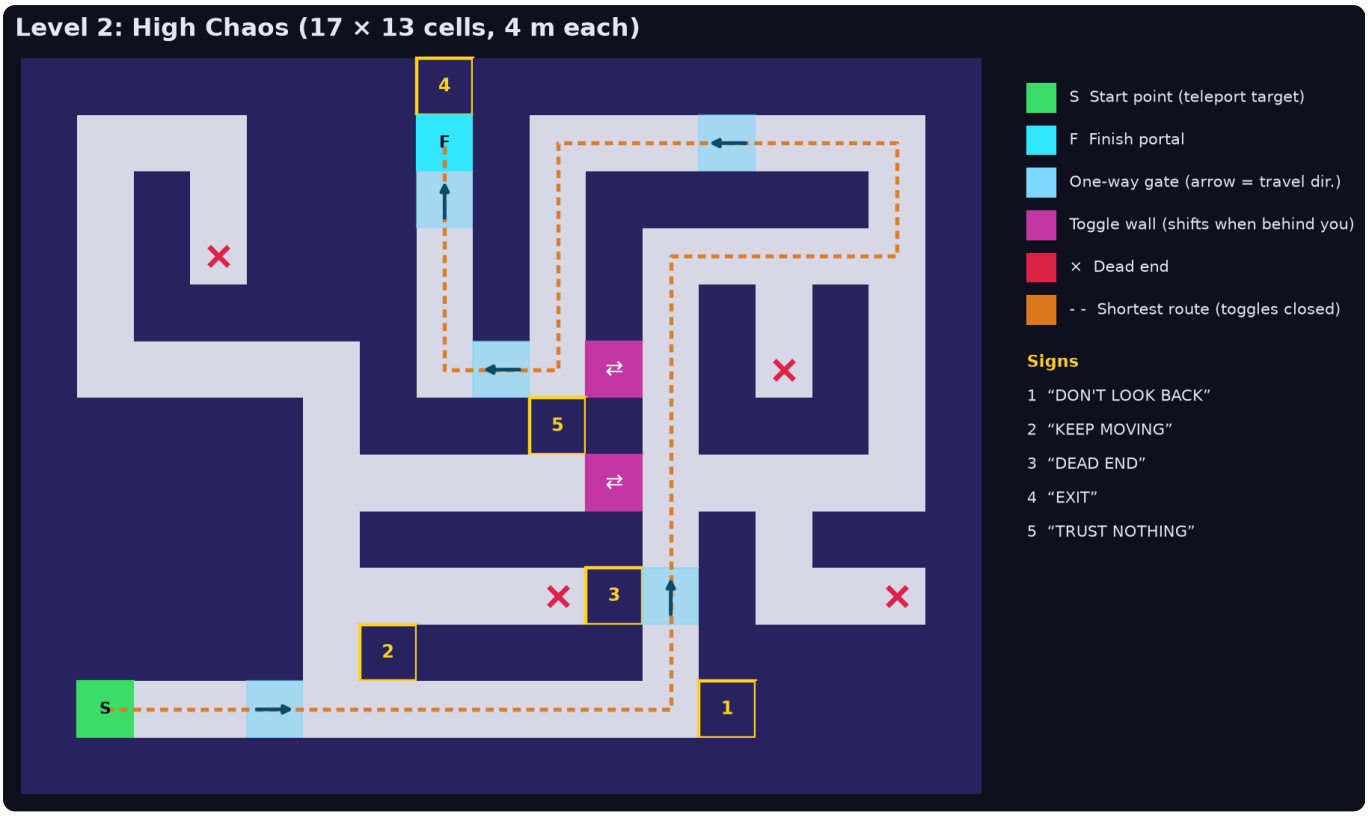
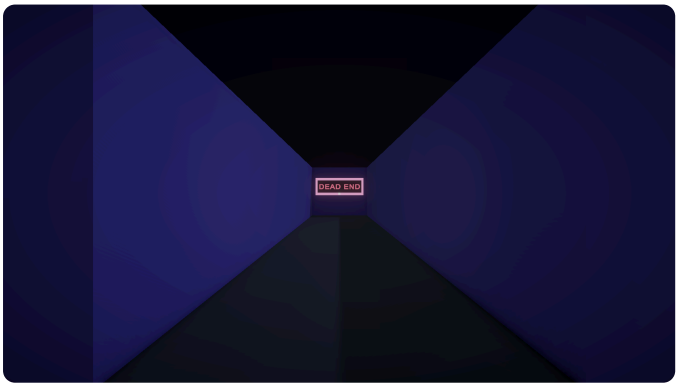
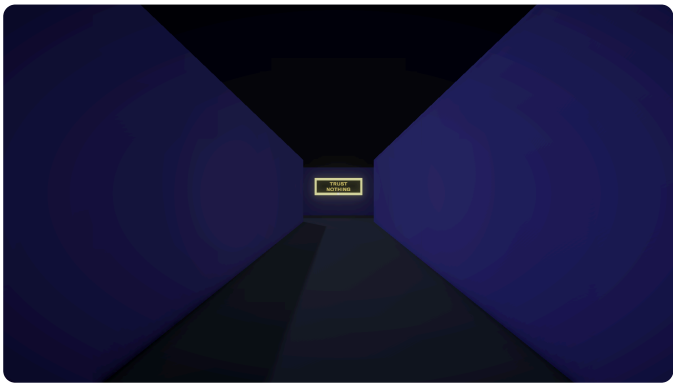


Figure 4. Level 2: 17 × 13 cells, 41-cell shortest route, 5 one-way gates, 2 fast toggle walls, 4 dead ends, and a large dead-end region on the left.

- **Trick corridor (sign 3, "DEAD END"):** leads towards the centre and stops exactly one wall short of the real path (gate `^` next to it). This teaches players not to trust where a corridor seems to lead.
- **Shortcut toggle (next to sign 5, "TRUST NOTHING"):** an open shortcut towards the exit that closes once it's behind you. If you leave it, you have to take the long way round.
- **Left region:** a tempting open area full of dead ends. It's reachable through a toggle wall that can close behind you.



The "DEAD END" trick corridor.



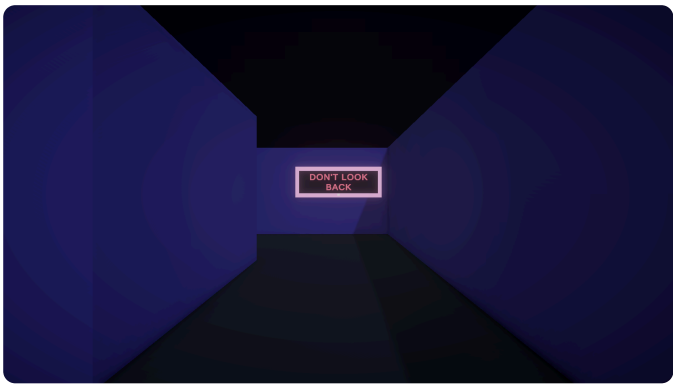
"TRUST NOTHING", looking down the central spine.

6. Visual style & feedback

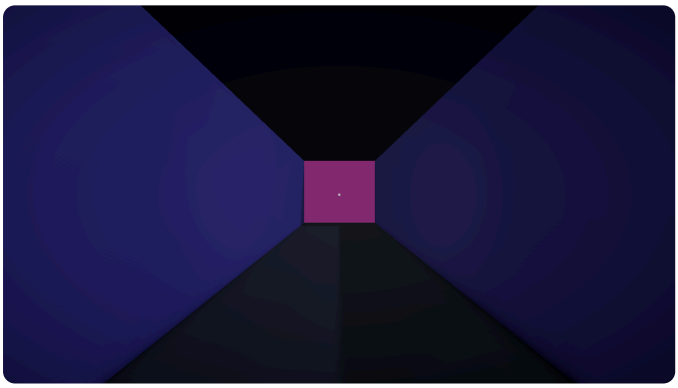
Dark neon toon. The world is built only from Unity primitive cubes, with flat colours and strong contrast: indigo walls, a near-black floor and dark fog. Anything important *glows* (bloom): the green START pad, the yellow, red and cyan signs, and the cyan portal. Purple marks the dynamic walls, so attentive players can spot them. A camera flashlight limits how far you can see, which adds to the tension.



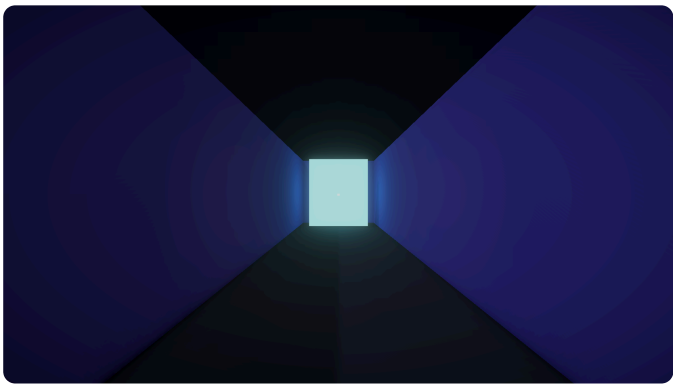
Stillness warning overlay with countdown (Rule 1 feedback).



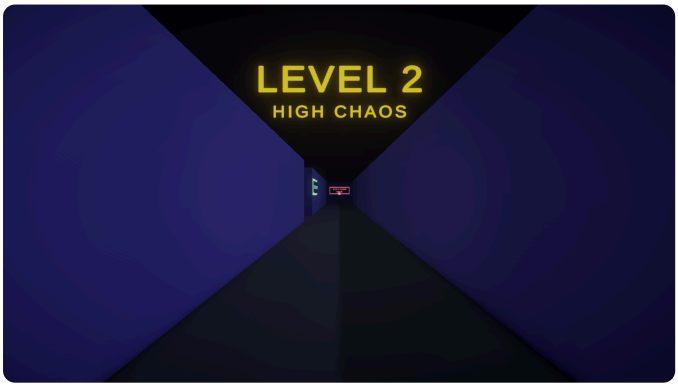
Glowing wall sign teaching Rule 2.



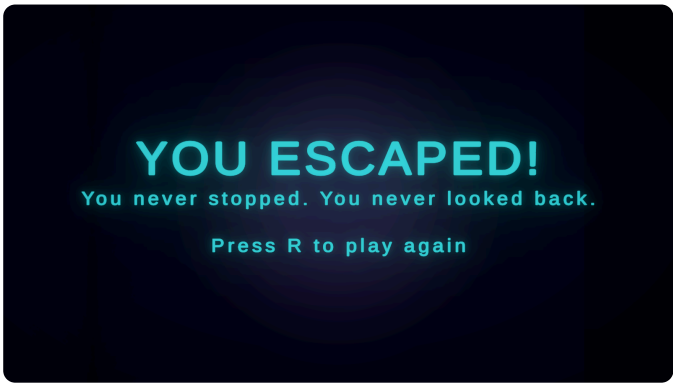
A one-way gate after it has closed behind the player.



The glowing Finish Portal.



Level intro message.



Victory screen.

Event	Feedback the player gets
Standing still	Red full-screen overlay fades in, with "DON'T STOP!" and the seconds left
Teleported	"TOO SLOW! BACK TO START" message, then back at the green START pad
Wall shifted	None at the moment it happens (it's always behind you). You discover it when you turn round. **Planned:** a "slam" sound.
Level start / complete	Large centre-screen title, then a fade

7. MDA analysis

Target aesthetics: **Sensation** and **Challenge**, with **Fantasy** (being hunted by a living maze) as flavour. The table follows three connected chains, each running from a mechanic to a dynamic to an aesthetic.

#	Mechanic	Dynamic	Aesthetic
1	**Stillness timer:** speed < 0.1 for 2 s means teleport to START. Countdown overlay.	Players never pause. They make split-second choices at junctions and "wiggle" to buy time, which still costs them.	**Urgency, adrenaline** (Sensation, Challenge)
2	**Line-of-sight wall shifting:** a wall is armed when seen and shifts once it's more than 110° behind you.	Players avoid turning round, glance back nervously, and stare at walls to "freeze" them.	**Paranoia, dread** (Sensation, Fantasy)
3	**Dead ends + one-way gates + stillness reset** (the reset also restores every wall).	Being trapped becomes a choice: players stand still *on purpose* to escape.	**Tension, then relief** (Challenge, Discovery)



Figure 5. The three MDA chains.

How the chains connect

Chain 1 feeds chain 2: you can't stop to watch your back, so the time pressure is what makes "don't look back" dangerous. **Chain 2 feeds chain 3:** walls closing behind you create real dead ends. **Chain 3 loops back to chain 1:** the way out of a trap is the stillness rule itself. One mechanic both punishes and rescues, and players learn to use it both ways.

> The Dynamics column describes the **intended** behaviour. It will be confirmed by the external playtest (Section 10). Chain 2 has already been revised once after the internal playtest showed that walls felt random rather than tense.

8. Design principles applied

Principle	Where it appears
Teach through level design	Level 1 is linear, and each sign sits right before the mechanic it describes. Level 2 only combines rules the player already knows.
Clear feedback	The stillness overlay has a visible countdown. Teleports always show a message. Every important object glows.
Signifiers / affordances	Dynamic walls are a different colour (purple) from static walls (indigo). The portal glows, which says "go here".
Difficulty curve	Level 1 has 2 gates and no branches. Level 2 has 5 gates, 2 toggle walls with a 3× faster cooldown, 4 dead ends and a trick corridor.
Meaningful choice / risk–reward	The toggle shortcut in Level 2 saves time but can close behind you.
Fairness & predictability	Walls follow one rule (seen, then behind you) and reset on teleport. The exit is always reachable, which was proven by a solver.

9. Technical implementation

Unity 6 (6000.4.0f1), Universal Render Pipeline, the new Input System and TextMeshPro. There are three runtime scripts with one-way dependencies: the player and the walls talk to `GameManager.Instance`.

Script	Responsibility
`FirstPersonController`	CharacterController movement with smoothed acceleration, smoothed mouse look with pitch clamp, the 2 s stillness timer, `TeleportToStart()`, and the Finish trigger (built-in `Finish` tag).
`DynamicMazeWall`	Each frame: frustum test + `Physics.Linecast` line-of-sight check, a "behind" angle test, cooldown and distance guards, then turns the Renderer and Collider on or off together. `ResetWall()` restores the starting state.
`GameManager`	Start Point reference, stillness overlay, centre messages, victory panel, wall reset on teleport, and level transitions (`nextSceneName`, empty = victory).
`SprintOrShiftLevelBuilder` (editor only)	Builds both scenes from text layouts: walls, gates, toggles, signs, START, portal, player, UI and lighting.

Why not `OnBecameInvisible()`? In the Editor it also counts the Scene-view camera, it stops firing once a wall hides its own Renderer, and it ignores what is in the way. Testing the player camera directly is the same idea, but predictable.

Hierarchy (identical in both levels): `Directional Light`, `Global Volume`, `GameManager`, `Environment` (Floor, StartPoint), `Maze` (Walls / DynamicWalls / Signs), `Player` → `PlayerCamera` → `Flashlight`, `UI` (Crosshair, StillnessOverlay, MessageText, VictoryPanel), `FinishPortal`.

10. Playtesting & iteration

Round 1: internal playtest (team, in the Editor)

Observation	Change
"The pink boxes for the dead ends are always there."	Debug gizmos were drawn in the Game view. They now draw only when a wall is selected. **Fixed**
"Moving the mouse is really slow."	Sensitivity raised from 0.12 to 0.45 and smoothing from 20 to 30. **Fixed**
"The purple walls seem random. Sometimes they're up before I've looked back."	Added line of sight and a "behind you" angle, and all walls now start open. **Fixed**. This was the most important change: it moved chain 2 from *confusion* to *paranoia*.

Round 2: automated rule checks

Scripted play-mode tests passed for: stillness teleport, gate closes behind the player, walls reset on teleport, walls hidden behind other walls don't count as seen, turning 90° to the side does not trigger a shift, turning fully round does, and the portal loads Level 2. A solver confirmed both mazes can be completed.

Round 3: external playtest

> **Not yet conducted.** Plan: 3 to 5 new players, controls only, about 10 minutes each. Tally junction pauses, turn-arounds and intentional resets (one per MDA chain), then a 5-question interview. The full script and note sheet are in the Playtesting Notes document.

11. Scope: implemented vs. planned

Feature	Status
First-person WASD movement + smooth mouse look	**Implemented**
Rule 1: 2 s stillness teleport + warning overlay	**Implemented**
Rule 2: line-of-sight wall shifting (gates + toggle walls)	**Implemented**
Rule 3: dead ends + wall reset on teleport	**Implemented**
Level 1 (tutorial) and Level 2 (high chaos), portal, transitions, victory screen	**Implemented**
Glowing signs, bloom, fog, flashlight	**Implemented**
Playable macOS build	**Implemented**
Audio: heartbeat during countdown, wall "slam", music	**Planned**
Pause menu with mouse-sensitivity slider; main menu	**Planned**
Telegraphing when a wall becomes "armed" (flicker / particles)	**Planned**
Run timer + best time per level	**Planned**
Level 3 (1.5 s stillness limit, mixed wall types)	**Planned**
Gamepad support, Windows build	**Planned**

12. Credits & references

- **Unity 6 / URP / Input System / TextMeshPro**: Unity Technologies. The Liberation Sans font (TMP Essential Resources) is under the SIL Open Font License 1.1.
- **No third-party art or audio.** All visuals are Unity primitives with project-made materials. Diagrams were made with Python/matplotlib.
- **Code:** written for this project with AI assistance from Claude Code (Anthropic). See the Team Contribution Summary.
- **Design references:** Hunicke, LeBlanc & Zubek (2004), *MDA: A Formal Approach to Game Design and Game Research*. Inspired by Red Light, Green Light (Statues) and the Weeping Angels from *Doctor Who*.
- **Repository:** https://github.com/adel-nabih/SprintOrShift